Braden McDorman

San Francisco Bay Area

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Summary

Engineering leader and systems builder with over a decade of experience in **simulation, robotics, and AI/ML tooling**. Built a robotics software company from scratch and now lead the platform team at Zoox enabling millions of autonomous vehicle simulations used to validate vehicle safety at scale.

Experience

Zoox – Engineering Manager – Foster City, CA — **Sep 2024 to Present**

Managing a six-person platform team of software and ML engineers whose tooling enables **hundreds of engineers across dozens of teams** to author and generate the millions of simulations used to validate Zoox's autonomous vehicle.

Defined and drove the team's new charter in generative AI, hiring all but two current members — including every ML engineer — to build out the capability from scratch.

Shipped agentic tooling that automated manual simulation workflows and deployed in-house generative models to production for exploratory vehicle testing.

Leading high-visibility cross-functional initiatives, partnering with Safety, Data Science, and ML Research under VP-level sponsorship and presenting regular updates directly to the CEO and CTO, to scale generative simulation as the foundation for AV validation at company scale.

Zoox – Senior Software Engineer – Foster City, CA — **Jun 2023 to Sep 2024**

Designed and built a 3D visual editor for authoring simulation scenarios, used across Zoox's engineering teams to author **thousands of scenarios** validating vehicle safety and testing autonomous vehicle components.

Added support for new Operational Design Domain (ODD) features to unblock upcoming company milestones, enabling engineers to author and validate scenarios for newly-supported driving capabilities ahead of launch.

Integrated an **interpretable traffic-infilling diffusion model** into the editor, giving engineers controllable generation of realistic surrounding traffic while authoring scenarios.

Semio – Co-Founder & CTO – Los Angeles, CA — **Jan 2015 to Apr 2023**

Co-founded Semio to build the OS, app ecosystem, and developer tools for social robots, growing the company to **$1M in annual revenue, entirely bootstrapped**.

Architected and built all aspects of Semio's products: web front-end, back-end, cloud infrastructure, 3D simulation, social behavior models, and on-robot software. Managed contractors and interns. Aligned and iterated with Fortune 100 customer teams to address their needs.

Learned how to (and *frequently how not to*) build a company and product from scratch.

KIPR – Lead Software Engineer – Norman, OK — **Jan 2015 to Jan 2016**

Led the design, development, and testing of the Wallaby robot controller for the Botball educational robotics program. This controller targeted novice K-12 and higher education students and is in use internationally by **tens of thousands of students and educators**.

Disney Research – Associate – Pittsburgh, PA **May 2015 to Aug 2015**
Implemented and conducted studies on group formation recognition and planning algorithms for social robots in human spaces. Published in the Proceedings of the Human Robot Interaction 2017 conference.

Earlier Experience

Microsoft · Software Development Engineer Intern · Redmond, WA — May 2014 to Aug 2014
NewSpin 360 · Research and Development Software Engineer · Remote — Jan 2014 to Apr 2014
OU · Teaching Assistant · Norman, OK — Aug 2013 to Mar 2014
KIPR · Software Engineer · Norman, OK — Jun 2011 to Feb 2014
OU · Research Assistant · Norman, OK — Aug 2013 to Jan 2014
Plasma @ UMass · Google Summer of Code Student · Remote — Jun 2013 to Sep 2013
OU Health Sciences Center · Student Researcher · Oklahoma City, OK — Aug 2011 to May 2012

Advisory Experience

KISS Institute for Practical Robotics (KIPR) – Board Member – Remote **Apr 2021 to Present**
Helping guide the mission and executive direction of KIPR, a 501(c)(3) nonprofit that runs the international Botball Educational Robotics Program, Junior Botball Challenge, and other STEM programs for K-12 educators and students, with particular focus on underserved communities. Also chairing KIPR's Technical Advisory Committee (since May 2019), advising engineering staff on product development and technical direction.

Bobble Sports – Advisor – Remote **Jan 2020 to Apr 2022**
Advised company founders on Bobble Sports' web-based event interaction system, including web frontend, app development, cloud backend, and high performance audio synthesis system.

Education

University of Southern California – Ph.D. Program, Computer Science **Aug 2015 to Dec 2016**
National Science Foundation Graduate Research Fellow, pursuing a Ph.D. in Computer Science with a focus in social robotics under Prof. Maja Matarić. Left the program *without receiving a degree* to pursue Semio, a robotics startup I co-founded, full-time.

University of Oklahoma – B.S. in Computer Science **Aug 2012 to May 2015**
Received B.S. in Computer Science with minor in Mathematics. Instructor for the Computer Science Interview Preparation club. Selected participant in DARPA Virtual Robotics Challenge.

Skills

Languages C | C++ | JavaScript | TypeScript | Python | Rust
Robotics & Simulation Robot Operating System | OpenGL | Qt | protobuf | gRPC | CMake
Infrastructure Docker | Kubernetes | node.js | React | Git | Linux | macOS | Embedded Linux
ML & Generative AI CUDA | Agentic Workflows | Diffusion Models | GenAI
Leadership & Methods Technical Leadership | People Management | Agile | REST / APIs | Microservice Architectures